

Q.SPGD (2 Marks)

Q.1 Define Plug Setting Multiplier (PSM)

Ans. Plug Setting Multiplier is the ratio of actual current in the relay coil to the relay's pick-up current.

Formula:

$\text{PSM} = \text{Fault Current} / \text{Pick-up Current}$

It indicates how strongly the relay operates during a fault.

Q.2 What is Symmetrical Fault

Ans. A symmetrical fault is a type of fault in a power system where all three phases are equally affected and the system remains balanced.

- It usually occurs when all three phase conductors are short-circuited together (with or without ground).
- It produces the maximum fault current compared to other faults.
- This fault is rare but very severe, so it is important for system design and protection.

Q.3 Write the formula Percentage Reactance (in terms of Base kVA)

Ans. Percentage reactance is given by:

$$\%X = \frac{X \times \text{Base kVA}}{(V)^2} \times 100$$

Where:

- X = Reactance in ohms
- V = System voltage (in volts or kV)
- Base kVA = Selected base power of the system

👉 This formula is used to express reactance in percentage form for easier comparison in power system calculations.

Q.4 Define Restriking Voltage & Recovery Voltage

Ans . Restriking Voltage:

It is the transient voltage that appears across the contacts of a circuit breaker immediately after the arc is extinguished.

It can cause the arc to strike again if insulation is not sufficient.

Recovery Voltage:

It is the normal steady-state voltage that appears across the circuit breaker contacts after final arc extinction when the system returns to normal condition.

Q.5 Why Earth Wire is Provided in overhead transmission line

Ans. Earth wire (also called shield wire) is provided above the transmission conductors to protect the line from lightning strokes.

- It intercepts lightning and diverts the current safely to ground through towers.
- Prevents damage to conductors and insulation.
- Reduces chances of power interruptions and faults.

👉 In short, it improves the reliability and safety of the transmission system.

Q.

Q. 6 Define Fuse & write at least two desirable Characteristics of fuse

Ans. Fuse:

A fuse is a protective device used in electrical circuits that melts and breaks the circuit when excessive current flows, thereby protecting equipment from damage.

Desirable Characteristics:

- It should operate quickly under fault conditions.
- It should be reliable and accurate in operation.
- It should have proper current rating (not blow under normal conditions).
- It should be cheap and easy to replace.

Q.7 What is Fusing Factor

Ans. Fusing Factor is the ratio of the minimum current required to melt the fuse to the rated current of the fuse. 🖱️ **Formula:**

Fusing Factor = (Minimum fusing current) / (Rated current of fuse)

🖱️ Key point:

- It is always greater than 1
- Because a fuse should not melt at its normal rated current

Q.8 Name the type of relay used to protect the long distance transmission line.

Ans. For long-distance transmission lines, the commonly used relay is:

🖱️ **Distance relay (Impedance relay)**

It measures the impedance between the relay location and the fault point. Since impedance depends on line length, it can accurately detect faults even over long distances.

Q.9 What are the devices that commonly used for protection against lightning surges

Ans. Devices commonly used for protection against lightning surges are:

- **Lightning arrester** – Diverts high surge voltage safely to earth
- **Surge absorber** – Absorbs and reduces the energy of surges
- **Surge diverter** – Redirects surge currents away from equipment
- **Earth (ground) wires** – Protect transmission lines by intercepting lightning strokes

🖱️ These devices protect electrical equipment from damage caused by sudden high voltages due to lightning.

Q.10 Define Short Circuit KVA.

Ans. Short Circuit kVA is the apparent power in a power system during a fault (short circuit condition). **Definition:**

It is the product of system voltage and the short-circuit current at the point of fault. **Formula:** Short Circuit kVA = Voltage × Short Circuit Current

Q.11 What are the switchgear equipments?

Ans. Switchgear equipment are devices used in a power system for control, protection, and isolation of electrical circuits.

Common switchgear equipment:

- Circuit breakers – interrupt fault current
- Isolators (disconnectors) – isolate a part of the system
- Relays – detect faults and give tripping signal
- Fuses – protect against overcurrent
- Busbars – distribute power within the system

 In short, switchgear ensures safe operation and protection of electrical equipment.

Q.12 What are the causes of over voltages?

Ans. Causes of over voltages in a power system:

- Lightning strikes – sudden high voltage surges from atmosphere
- Switching operations – opening/closing of breakers or lines
- Insulation failure – breakdown can create abnormal voltage rise
- Sudden load changes – load removal can increase system voltage
- Resonance effect – interaction of inductance and capacitance
- Fault conditions – like earth fault or arcing ground

 These conditions cause voltage to rise above normal, which can damage equipment if not controlled.

Q.13 Define relay and circuit breaker.

Ans. Relay:

A relay is a protective device that detects faults in a power system and sends a signal to operate the circuit breaker.

Circuit Breaker:

A circuit breaker is a switching device that automatically interrupts current flow when a fault is detected, protecting the system from damage.

Q.14 What are the types of faults in a power system?

Ans. 1. Symmetrical Fault

- All three phases are equally affected
- Example: 3-phase fault (L-L-L)
- Rare but very severe

2. Unsymmetrical Faults

These are more common:

- Line to Ground (L-G) → one phase touches ground
- Line to Line (L-L) → two phases short together
- Double Line to Ground (L-L-G) → two phases + ground

👉 Key point:

- Unsymmetrical faults occur most frequently
- Symmetrical faults cause maximum damage but are less common

Q. 15 Define short-circuit kVA.

Ans. Short-circuit kVA is the apparent power in a power system under fault (short-circuit) conditions. **Definition:**

It is the product of system voltage and the short-circuit current at the point of fault. **Formula:**

Short-circuit kVA = Voltage × Short-circuit current

Q.16 Why do we use reactors in a power system?

Ans. Reactors are used in a power system mainly to control and improve system performance.

Uses of reactors:

- Limit fault current – reduces short-circuit current to protect equipment
- Control voltage – helps maintain proper voltage levels
- Improve system stability – stabilizes power flow in the system
- Reduce switching surges – absorbs sudden voltage changes

Q. 17 Write down the operating principle of circuit breaker.

Ans. A circuit breaker works on the principle of arc interruption.

👉 When a fault occurs:

- The relay detects the fault and sends a trip signal
- The circuit breaker opens its contacts
- An arc is formed between the contacts due to current flow
- The breaker uses a medium (air, oil, SF₆, or vacuum) to extinguish the arc
- Once the arc is extinguished, current flow stops, and the circuit is isolated

Q.18 Define pick-up current and plug setting multiplier.

Ans. Pick-up Current:

It is the minimum current at which a relay just starts to operate.

Plug Setting Multiplier (PSM):

It is the ratio of fault current to pick-up current. **Formula:**

$PSM = \text{Fault current} / \text{Pick-up current}$ **Key idea:**

- Pick-up current → starting point of relay operation
- PSM → tells how much the fault current exceeds that value

Q.19 What is surge absorber?

Ans. A surge absorber is a device used in power systems to absorb and reduce sudden high-voltage surges (like lightning or switching surges), protecting electrical equipment from damage.

Q.20 Write down the important faults in on an alternator.

Ans. Important faults in an alternator

1. Stator faults

- Stator winding earth fault
- Phase-to-phase fault
- Inter-turn fault in winding

2. Rotor faults

- Rotor earth fault
- Field winding short circuit
- Loss of excitation

3. Abnormal conditions

- Overloading
- Unbalanced loading
- Over-speed
- Cooling system failure